

INTERNATIONAL A-LEVEL PSYCHOLOGY

PS03

Unit 3 Advanced Topics and Research Methods 2

Mark scheme

June 2025

Version: 0.1 Pre-Standardisation



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from www.oxfordaqa.com

Copyright information

OxfordAQA retains the copyright on all its publications. However, registered schools/colleges for OxfordAQA are permitted to copy material from this booklet for their own internal use, with the following important exception: OxfordAQA cannot give permission to schools/colleges to photocopy any material that is acknowledged to a third party even for internal use within the centre.

Copyright © 2025 OxfordAQA International Examinations and its licensors. All rights reserved.

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Section A: Psychology of Sleep

Total for this section: 30 marks

Question	Marking Guidance	Total Marks						
01	There are three types of biological rhythm. A Circadian B Infradian C Ultradian Complete the table of definitions by choosing the correct letter from the list above and writing it next to the appropriate definition. Use each letter only once. <table><tr><td>Biological rhythm with a cycle length of 24 hours.</td><td>A (Circadian)</td></tr><tr><td>Biological rhythm with a cycle length of less than 24 hours.</td><td>C (Ultradian)</td></tr><tr><td>Biological rhythm with a cycle length of more than 24 hours.</td><td>B (Infradian)</td></tr></table>	Biological rhythm with a cycle length of 24 hours.	A (Circadian)	Biological rhythm with a cycle length of less than 24 hours.	C (Ultradian)	Biological rhythm with a cycle length of more than 24 hours.	B (Infradian)	3 AO1 = 3
Biological rhythm with a cycle length of 24 hours.	A (Circadian)							
Biological rhythm with a cycle length of less than 24 hours.	C (Ultradian)							
Biological rhythm with a cycle length of more than 24 hours.	B (Infradian)							

Question	Marking Guidance	Total Marks
02	Outline one limitation of research studies into circadian rhythms. Award marks as follows 2 marks for a clear outline of a limitation of research studies into circadian rhythms. 1 mark for a muddled or vague outline of a limitation of research studies into circadian rhythms. Possible content • Issues with ecological validity of studies such as Siffre, eg conducted in an underground cave so might tell us little about natural circadian rhythms. • Research often involves small and limited samples (lacks population validity), eg Siffre was the sole participant and male. • Possibility of extraneous variables in research due to difficulty in isolating circadian biological processes from other biological processes. Credit other relevant material.	2 AO3 = 2

03	Omar has just started his new job as a pilot, flying around the world. He notices that when he flies in the direction of west to east, he experiences more severe jet lag than when he flies from east to west.	
----	---	--

Question	Marking Guidance	Total Marks												
03.1	Using your knowledge of the effects of jet lag on the disruption of biological rhythms, explain Omar's experiences. Possible content - When Omar flies from east to west his body clock/circadian rhythm runs ahead of the new time zone (phase delay). Consequently, Omar may simply stay awake longer in the new time zone to quickly adjust his body clock/circadian rhythm to the new local time. Therefore, the effects of jet lag will be less severe. - When Omar flies from west to east his body clock/circadian rhythm runs behind the new time zone (phase advance). As a result, Omar's body clock/circadian rhythm needs to advance/move forward to catch up to the new time zone. Omar's body clock/circadian rhythm will find it much harder and take longer to make this adjustment. Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Application of the effects of jet lag on Omar are accurate with some detail. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Application of the effects of jet lag on Omar are limited, vague or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Application of the effects of jet lag on Omar are accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4	1	Application of the effects of jet lag on Omar are limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO2 = 4
Level	Description	Marks												
2	Application of the effects of jet lag on Omar are accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4												
1	Application of the effects of jet lag on Omar are limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2												
0	No creditable content.	0												

Question	Marking Guidance	Total Marks
3.2	Another pilot suggests to Omar that he should avoid consuming any coffee before travelling, as this might make jet lag symptoms worse. Suggest two other behaviours that Omar could change to reduce the effects of jet lag when travelling from west to east. For each suggested behaviour, award marks as follows 2 marks for a clear suggestion of a behaviour that Omar could use to reduce the effects of jet lag. 1 mark for a muddled or vague suggestion of a behaviour that Omar could use to reduce the effects of jet lag. Possible content • Omar could try to adjust his daily schedule (eg sleeping and eating) before flying in anticipation of the new time zone. • Omar could try to sleep well before he flies to avoid starting in a state of sleep-deprivation. • Omar could avoid consuming any alcohol as this has been shown to make jet lag symptoms worse. • Omar could use melatonin supplements to help him sleep at the appropriate local time. • Omar should expose himself to morning daylight as soon as possible when arriving in the new time zone. Credit other relevant material.	4 AO2 = 4

Question	Marking Guidance	Total Marks
04	Name one type of sleep. Award 1 mark for naming either non-REM/NREM or REM sleep. Note: Credit slow-wave sleep (SWS) as an alternative name for non-REM. Note: No credit for saying light or deep sleep.	1 AO1 = 1

Question	Marking Guidance	Total Marks												
05	Evaluate the role of personality factors as an explanation for insomnia. Possible content • Use of supporting evidence for the role of anxiety/anxious personalities and insomnia, eg Vahtera et al. (2007), Gregory et al. (2006), Heath et al. (1998); and for association with mental illness and insomnia, eg Morin et al. (1999). • Use of therapies with people with anxious personalities, such as CBT, can reduce arousal and anxiety levels and have been shown to be effective in helping insomnia. • Methodological issues of associated research studies, eg limited sample sizes, issues with extraneous variables, use of correlation research. These can reduce the validity of the explanation. Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>2</td><td>Evaluation of the role of personality factors as an explanation for insomnia is accurate with some detail. The answer is clear with appropriate use of specialist terminology.</td><td>3–4</td></tr> <tr> <td>1</td><td>Evaluation of the role of personality factors as an explanation for insomnia is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	2	Evaluation of the role of personality factors as an explanation for insomnia is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4	1	Evaluation of the role of personality factors as an explanation for insomnia is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	4 AO3 = 4
Level	Description	Marks												
2	Evaluation of the role of personality factors as an explanation for insomnia is accurate with some detail. The answer is clear with appropriate use of specialist terminology.	3–4												
1	Evaluation of the role of personality factors as an explanation for insomnia is limited, vague or muddled. Specialist terminology is either absent or inappropriately used.	1–2												
0	No creditable content.	0												

Question	Marking Guidance	Total Marks									
06	Discuss memory consolidation as a function of sleep. Possible knowledge • Sleep is thought to be a time when memories acquired during the day are changed and established into permanent memories. • REM sleep is believed to have a role in the consolidation of procedural memories. • NREM (particularly the deeper stages) is linked to consolidation of declarative (episodic and semantic) memories. • During NREM, declarative memories are reactivated and then redistributed to the frontal cortex by the hippocampus. • Description of methodology of how memory consolidation is researched, eg use of sleep laboratories and brain scanning methods such as EEG. • Description of specific research studies investigating memory consolidation, eg Karni et al. (1994), Walker et al. (2002), Diekelmann & Born (2010). Possible discussion • Use of research studies to support memory consolidation as a function of sleep, eg Karni et al. (1994), Walker et al. (2002), Diekelmann & Born (2010). • Advances in techniques used to measure brain activity have helped to improve our understanding of the function of sleep in memory consolidation. • Methods used to measure brain activity can lack precision and therefore do not give a full and/or detailed understanding of the function of sleep in memory consolidation. • The function of sleep in memory consolidation is very complex and, as of yet, we do not fully understand all of the processes involved. • Issues with extraneous variables in research, eg in early studies it was often difficult to distinguish between the function of sleep and simply being quiet, relaxed and undisturbed. Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>4</td><td>Knowledge of memory consolidation as a function of sleep is accurate and generally well detailed. Discussion is effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.</td><td>10–12</td></tr> <tr> <td>3</td><td>Knowledge of memory consolidation as a function of sleep is evident but there are occasional inaccuracies and/or omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is used appropriately.</td><td>7–9</td></tr> </tbody> </table>	Level	Description	Marks	4	Knowledge of memory consolidation as a function of sleep is accurate and generally well detailed. Discussion is effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	10–12	3	Knowledge of memory consolidation as a function of sleep is evident but there are occasional inaccuracies and/or omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is used appropriately.	7–9	12 AO1 = 6 AO3 = 6
Level	Description	Marks									
4	Knowledge of memory consolidation as a function of sleep is accurate and generally well detailed. Discussion is effective. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	10–12									
3	Knowledge of memory consolidation as a function of sleep is evident but there are occasional inaccuracies and/or omissions. There is some effective discussion. The answer is mostly clear and organised but occasionally lacks focus. Specialist terminology is used appropriately.	7–9									

	2	Limited knowledge of memory consolidation as a function of sleep is present. Any discussion is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately.	4–6	
	1	Knowledge of memory consolidation as a function of sleep is very limited. Discussion is limited, poorly focussed or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used.	1–3	
	0	No creditable content.	0	

Section B: Schizophrenia

Total for this section: 30 marks

Question	Marking Guidance	Total Marks
07	What is meant by positive symptoms and negative symptoms of schizophrenia? Award 1 mark for each correct definition • Positive symptoms are (atypical) symptoms which are experienced in addition to normal functioning. • Negative symptoms are (atypical) symptoms that represent the loss of usual or expected functioning. Credit alternative wording. Note: No credit for simply naming examples of positive and/or negative symptoms of schizophrenia.	2 AO1 = 2

Question	Marking Guidance	Total Marks
08	What is meant by hallucinations? Award marks as follows 2 marks for a clear definition of hallucinations. 1 mark for a muddled or vague definition of hallucinations. Possible content • Hallucinations are a false perception of objects or events involving the senses, these often include seeing and/or hearing things that are not real. Credit alternative wording.	2 AO1 = 2

Question	Marking Guidance	Total Marks															
09	Evaluate the use of anti-psychotic drugs as a therapy for schizophrenia Possible content - Side effects can cause high attrition rates, eg typical anti-psychotic drugs can cause confusion, weight gain, involuntary movement, so some people might stop taking them. - Use of evidence for effectiveness of anti-psychotic drugs, eg Davies et al. (1990), Thornley (2003), Adams et al. (2005), Cole et al. (1964), Meltzer (1999), McGlashan et al. (2006). - Typical anti-psychotics have no/little effect on negative symptoms however atypical can help with positive and negative symptoms. - Not all patients respond well to anti-psychotic drugs. - Anti-psychotic drugs only treat the symptoms and do not offer a cure and therefore must be taken regularly even when symptoms are not currently being experienced. - Anti-psychotic drugs are accessible for patients as many of them can be taken in a variety of forms such as injectables or syrups. - Typical anti-psychotics are relatively cheap but some of the newer atypical drugs are very expensive. Credit other relevant material. <table border="1"> <thead> <tr> <th>Level</th><th>Description</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>3</td><td>Evaluation of the use of anti-psychotic drugs is detailed and effective. The answer is clear with appropriate use of specialist terminology.</td><td>5–6</td></tr> <tr> <td>2</td><td>Evaluation of the use of anti-psychotic drugs is relevant but effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.</td><td>3–4</td></tr> <tr> <td>1</td><td>Evaluation of the use of anti-psychotic drugs is very limited. The answer is vague and/or muddled. Specialist terminology is either absent or inappropriately used.</td><td>1–2</td></tr> <tr> <td>0</td><td>No creditable content.</td><td>0</td></tr> </tbody> </table>	Level	Description	Marks	3	Evaluation of the use of anti-psychotic drugs is detailed and effective. The answer is clear with appropriate use of specialist terminology.	5–6	2	Evaluation of the use of anti-psychotic drugs is relevant but effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4	1	Evaluation of the use of anti-psychotic drugs is very limited. The answer is vague and/or muddled. Specialist terminology is either absent or inappropriately used.	1–2	0	No creditable content.	0	6 AO3 = 6
Level	Description	Marks															
3	Evaluation of the use of anti-psychotic drugs is detailed and effective. The answer is clear with appropriate use of specialist terminology.	5–6															
2	Evaluation of the use of anti-psychotic drugs is relevant but effectiveness is lacking. The answer lacks clarity in places. Specialist terminology is occasionally used appropriately.	3–4															
1	Evaluation of the use of anti-psychotic drugs is very limited. The answer is vague and/or muddled. Specialist terminology is either absent or inappropriately used.	1–2															
0	No creditable content.	0															

Question	Marking Guidance	Total Marks
10	Sarah and George are reading about schizophrenia in their psychology textbooks. Sarah says, “I thought that you could only develop schizophrenia if someone in your family already had it. My friend has schizophrenia and so does her biological father.” George replies, “I thought that schizophrenia was related to something that was happening inside a person’s brain.” Describe and evaluate two biological explanations of schizophrenia. Refer to the conversation between Sarah and George in your answer. Possible description Genetics • Schizophrenia could be hereditary. • Thought to be polygenic as several candidate genes (eg COMT gene) have been associated with increased risk, such as those genes that code for production and function of neurotransmitters such as dopamine. • Description of how family, twin and adoption studies are used to investigate the genetic explanation. • Description of family, twin and adoption studies, eg Gottesman & Shields (1966 and 1982), Tienari et al. (1994). Neural Correlates • Dopamine hypothesis suggests that reduced dopamine in the frontal area of the brain causes negative symptoms whilst increased dopamine in other areas is responsible for positive symptoms. • Glutamate hypothesis suggests that glutamate activity is reduced in people with schizophrenia, linked to the NMDA receptors working less effectively. Glutamate acts to reduce dopamine, therefore if glutamate is reduced, then this increases dopamine levels. • Changes in the structure of the brain might be correlated to behavioural changes seen in schizophrenia, eg a negative correlation between activity levels in the ventral striatum and the severity of overall negative symptoms, reduced activity in the superior temporal gyrus and the anterior cingulate gyrus has been shown to correlate with positive symptoms. • Description of research into neural correlates, eg Johnstone et al. (1976). Possible application • Sarah is highlighting the genetic explanation of schizophrenia. • She suggests it runs in families, believing that a gene or multiple genes are inherited and passed on. • She states that her friend has schizophrenia and so does her friend’s biological father, suggesting that a gene(s) for schizophrenia may have been passed down. • George is suggesting the neural correlates explanations of schizophrenia. • He believes that it is a change within the brain. George is therefore highlighting a potential issue with neurotransmitters in the brain such as dopamine or changes in brain (ventricle) structure/functioning.	20 AO1 = 8 AO2 = 4 AO3 = 8

Possible evaluation

- Use of supporting evidence, eg Gottesman & Shields (1966 and 1982), Kety et al. (1968), Tienari et al. (1994), Ripke et al. (2014), Kim et al. (1980), Juckel et al. (2006).
- Methodological issues with research when clearly linked to biological explanations for schizophrenia, eg twin studies have used small samples and evidence suggests that MZ twins are reared in a more similar environment than DZ twins, and therefore nurture might partly explain the higher concordance rate found for MZ twins rather than just nature.
- Anti-psychotic drugs which block dopamine receptors have been shown to help to reduce symptoms.
- Oversimplification of the dopamine hypothesis as many neurotransmitters interact and singling out just one neurotransmitter might be inaccurate.
- Issue with unreliable findings of research, eg Lewis (1990) conducted a meta-analysis on 18 studies and found no significant link between enlarged ventricles and negative symptoms associated with schizophrenia.
- A variety of brain mechanisms is likely to be implicated in schizophrenia, but none are yet sufficiently understood to provide a causal link.
- Explanations are reductionist as environmental risk factors are implicated in the onset of schizophrenia and it is likely that biological factors only contribute partially towards the development of schizophrenia (interactionist approach).

Credit other relevant material.

Level	Description	Marks
4	Knowledge of two biological explanations of schizophrenia is accurate and generally well detailed. Evaluation is effective. Application is appropriate. Minor detail and/or expansion of argument is sometimes lacking. The answer is clear and focused. Specialist terminology is used effectively.	16–20
3	Knowledge of two biological explanations of schizophrenia is evident but there are occasional inaccuracies and/or omissions. There is some effective evaluation. There is some appropriate application. The answer is clear and organised but occasionally lacks focus. Specialist terminology is used appropriately. OR one biological explanation at Level 4.	11–15
2	Limited knowledge of two biological explanations of schizophrenia is present. Any evaluation/application is of limited effectiveness. The answer lacks clarity, accuracy and organisation in places. Specialist terminology is occasionally used appropriately. OR one biological explanation at Level 3.	6–10

	1	Knowledge of two biological explanations of schizophrenia is very limited. Evaluation/application is limited, poorly focused or absent. The answer as a whole lacks clarity, has many inaccuracies and is poorly organised. Specialist terminology is either absent or inappropriately used. OR one biological explanation at Level 2 or 1.	1–5	
	0	No creditable content.	0	

Section C: Research Methods 2

Total for this section: 30 marks

	A psychologist was interested in studying the social behaviours people show towards each other in public. The psychologist decided to conduct an observation. To carry out this observation, the psychologist visited a large shopping centre and sat in a busy area. The psychologist observed and recorded some of the social behaviours that the people in the shopping centre showed towards each other. The people in the shopping centre did not know that they were being observed as part of the study.	
--	--	--

Question	Marking Guidance	Total Marks
11	Which one of the following set of observational techniques is being used by the psychologist in this study? Shade one box only. Answer key: C – Covert, naturalistic and non-participant observation	1 AO2 = 1

Question	Marking Guidance	Total Marks
12	Identify the sampling technique used by the psychologist in this study. Justify your answer. Award 1 mark for opportunity sampling. Award 1 further mark for stating that the psychologist was studying/observing whoever was in the shopping centre at the time of the observation. Accept alternative wording.	2 AO2 = 2

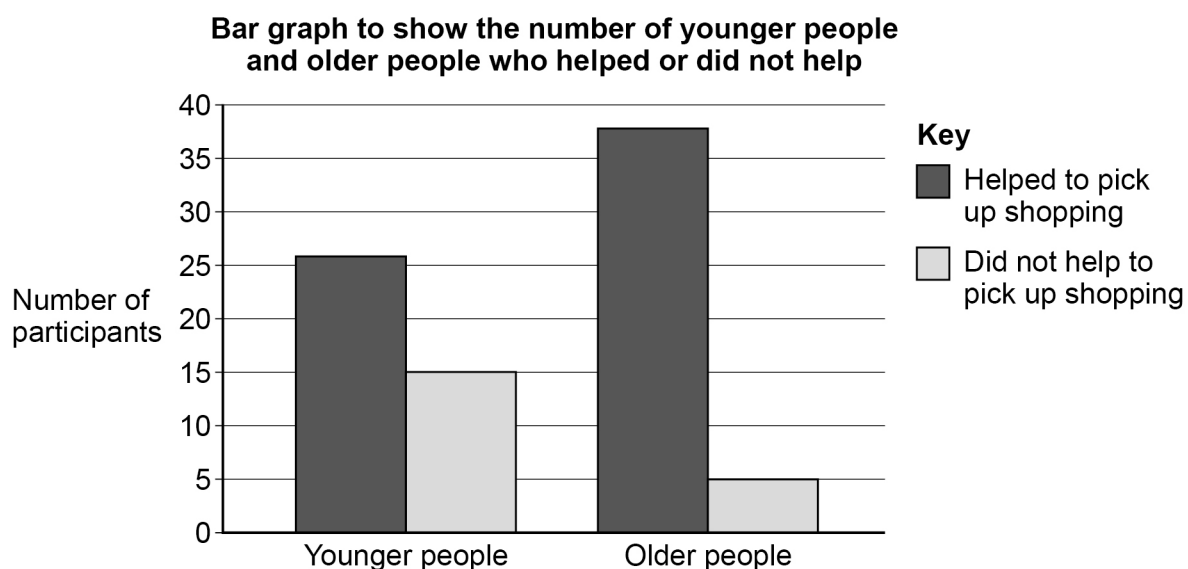
Question	Marking Guidance	Total Marks
13	One of the psychologist's behavioural categories was 'holding a door open for another person'. Suggest two other behavioural categories that the psychologist might record during the observation of social behaviours. Award 1 mark each for any two behavioural categories that would be appropriate for an observation on social behaviours. Possible content • Talking (to another person). • Laughing (with another person). • Making eye contact. • Eating a meal (with someone). • Helping someone who has dropped something. • Helping someone who has fallen. • Apologising (for bumping into someone). • Smiling (at another person). Credit other relevant material.	2 AO2 = 2

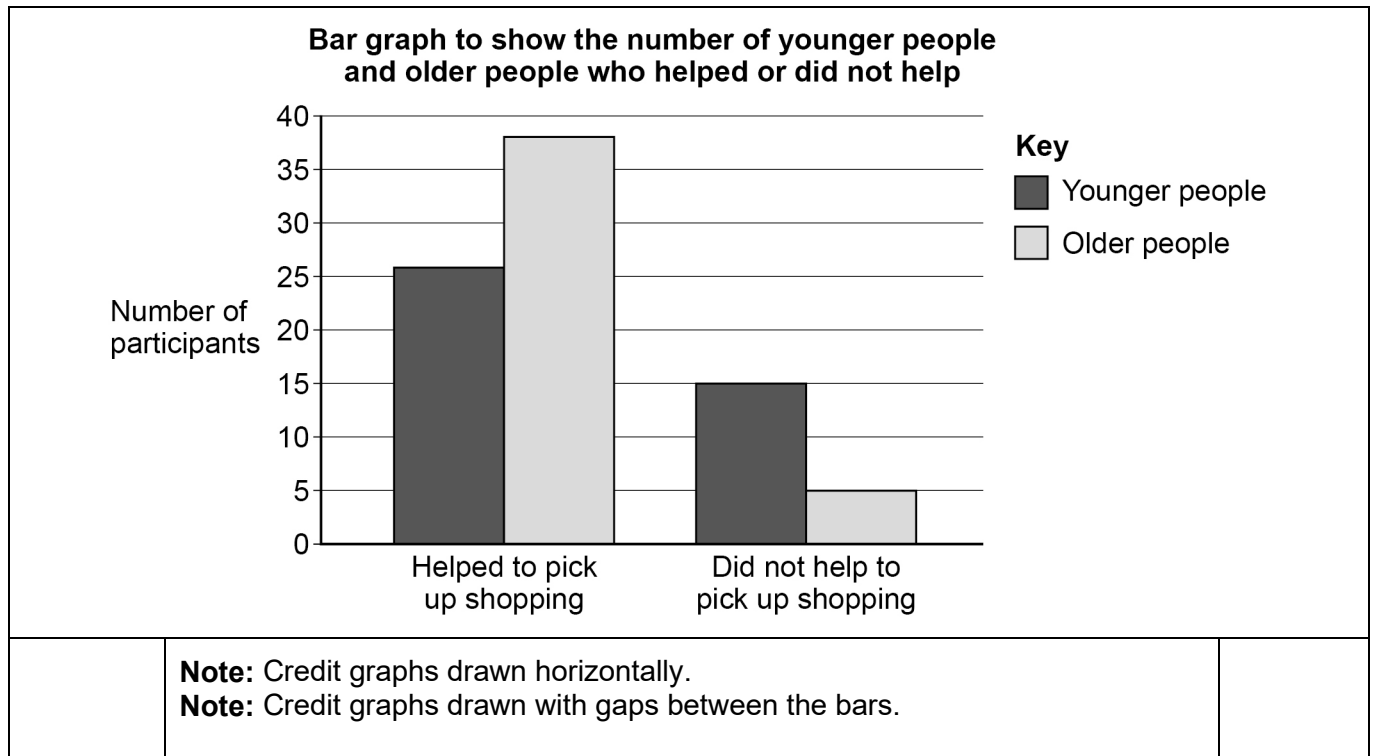
Question	Marking Guidance	Total Marks
14	The psychologist believed that the observation had a high level of ecological validity. Explain why the psychologist felt confident about the ecological validity of the observation. Award marks as follows 3 marks: The explanation of why the psychologist felt confident about the ecological validity of the observation is detailed. The answer is clear with appropriate use of specialist terminology. 2 marks: The explanation of why the psychologist felt confident about the ecological validity of the observation lacks detail. The answer lacks clarity in places. 1 mark: The explanation of why the psychologist felt confident about the ecological validity of the observation is briefly presented. The answer is very limited, vague and/or muddled. Possible content • The observation was carried out in the real-life setting of a shopping centre with people undertaking their own day-to-day activities. • The people in the shopping centre were unaware that they were being observed and therefore would have acted naturally. • The psychologist should feel confident that the behaviours observed reflected natural behaviours that the people would have shown even if the observation was not taking place. Credit other relevant material.	3 AO2 = 3

	During the observation, the psychologist noticed differences in the behaviours of younger people and older people. One main difference was related to helping behaviour. The psychologist decided to conduct a field experiment to investigate the differences in helping behaviour between younger people and older people. To conduct this field experiment, the psychologist returned to the shopping centre on a different day with an assistant. The assistant dropped bags of shopping in front of either younger people or older people. The psychologist recorded whether the younger people or the older people helped to pick up the shopping or not.	
--	---	--

Question	Marking Guidance	Total Marks
15	What is a field experiment? Award marks as follows 2 marks for a clear and relevant outline of a field experiment. 1 mark for a muddled or vague outline of a field experiment. Possible content • A (controlled) experimental study conducted in a more natural/everyday setting where the researcher directly manipulates an IV and measures its effects on the DV. • Participants often do not know that they are taking part in a field experiment. Credit alternative wording.	2 AO1 = 2

Question	Marking Guidance	Total Marks									
16	The results of the field experiment are shown in Table 1 below. Table 1: The helping behaviour shown by younger people and older people. <table border="1"> <thead> <tr> <th></th><th>Younger people</th><th>Older people</th></tr> </thead> <tbody> <tr> <td>Helped pick up shopping</td><td>26</td><td>38</td></tr> <tr> <td>Did not help pick up shopping</td><td>15</td><td>5</td></tr> </tbody> </table> Draw an appropriate graph to show the results of this field experiment. Your graph must include a title. Award marks as follows 1 mark for an accurately drawn bar graph. 1 mark for an appropriately labelled x-axis. 1 mark for an appropriately labelled y-axis. 1 mark for an appropriate graph title. 1 mark for a key/labelling of the bars.		Younger people	Older people	Helped pick up shopping	26	38	Did not help pick up shopping	15	5	5 AO2 = 5
	Younger people	Older people									
Helped pick up shopping	26	38									
Did not help pick up shopping	15	5									





Question	Marking Guidance	Total Marks
17	The psychologist collected quantitative data in this field experiment. Outline what is meant by quantitative data? Award marks as follows 2 marks for a clear and relevant outline of quantitative data. 1 mark for a muddled or vague outline of quantitative data. Possible content Data that is numerical such as scores from tests or tallies of yes/no answers. Credit alternative wording.	2 AO1 = 2

Question	Marking Guidance	Total Marks
18	Explain one benefit of collecting quantitative data in psychological research. Award marks as follows 2 marks for a clear and relevant benefit of collecting quantitative data. 1 mark for a muddled or vague benefit of collecting quantitative data. Possible content - Quantitative data can be easier to analyse (compared to qualitative data) as you can put the numerical data into graphs and conduct statistical tests on the data which makes it easier to draw conclusions. - Minimises the chances of researcher bias as quantitative data is objective and is therefore not open to subjective interpretation (like qualitative data). - Quantitative data allows for a greater level of generalisation (compared to qualitative data) as it is usually obtained from a larger sample and analysed as a whole. Credit other relevant material.	2 AO3 = 2

Question	Marking Guidance	Total Marks
19	Identify a suitable statistical test that the psychologist could use to analyse the data from the field experiment. Explain two reasons for your choice. Award marks as follows 1 mark for identifying that a Chi-squared test would be the most suitable. and For any two of the following award 2 marks for a clear and coherent reason explicitly linked to the study. 1 mark for a limited reason implicitly linked to the study. Possible content • The psychologist is testing to see if there is a difference between younger people and older people in relation to their helping behaviour. • The psychologist is using an independent group design as participants are either younger people or older people. • The data collected by the psychologist are treated as nominal as it is a discrete data that is separated into separate categories of helping (to pick up shopping) or not helping (to pick up shopping). Note: credit answers that refer to looking for an association between two variables (age and helping behaviour) rather than a difference. Note: appropriate reasons can still be credited even if an incorrect test is named or no test is given. Note: where more than two reasons are given, only the first two should be credited.	5 AO2 = 5

Question	Marking Guidance	Total Marks
20	Outline two limitations of conducting field experiments. For each limitation, award marks as follows 2 marks for a clear and relevant limitation of a field experiment. 1 mark for a muddled or vague limitation of a field experiment. Possible content • Less control over extraneous variables so it is harder to establish cause and effect. • Can be harder to replicate in exactly the same way due to the lower levels of control so reliability can be harder to assess. • Potential of ethical issues as participants are being studied without giving their consent (which is often obtained afterwards) which might upset some participants. Credit other relevant material.	4 AO3 = 4

Question	Marking Guidance	Total Marks
21	After conducting psychological research, psychologists often write a scientific report about their investigation for other people to read. Briefly outline two pieces of information a psychologist would include in the Introduction section of a scientific report. Award 1 mark for any two of the following • A background of the study/rationale for the current study. • Reviews/summaries of past relevant research related to the current study (literature review). • The aim of the current study. • The hypotheses of the current study.	2 AO1 = 2

PS03 grid

	AO1	AO2	AO3	Total
Section A				
01	3			3
02			2	2
03.1		4		4
03.2		4		4
04	1			1
05			4	4
06	6		6	12
Section B				
07	2			2
08	2			2
09			6	6
10	8	4	8	20
Section C				
11		1		1
12		2		2
13		2		2
14		3		3
15	2			2
16		5		5
17	2			2
18			2	2
19		5		5
20			4	4
21	2			2
Unit total	28	30	32	90